

Perception of Difficulty Changes in Sign Language: The Influence of Middle School Students' Sign Language Cognition and Basic Linguistic Knowledge

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Table of Contents

Abstract	1
CHAPTER1 Introduction	2
CHAPTER 2 Literature Review	4
2.1 What is Sign Language Linguistics?	4
2.2 The role of online platforms in the dissemination of sign language	6
2.3 Research evidence on the cognitive impacts of sign language on students	7
CHAPTER 3 Methodology	8
3.1 Research questions	8
3.2 Method of data collection	8
3.3 Research participants	10
3.4 Data analysis	10
3.5 Ethical considerations	11
CHAPTER 4 Findings	12
4.1 The Constraints of Digital Media in Sign Language Dissemination	12
4.2 The Innate Logical Aptitude of High School Students for Sign Language Acquisition.	15
4.3 An Overall Increase in Students' Acceptance of Sign Language Following the Acquisition of Foundational Linguistic Knowledge	19
4.4 The Demand for Credible and Formalized Learning Channels Among Non-Signers	21
CHAPTER5 Conclusion	23
References	25
Appendix	27

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Abstract

This study examines changes in secondary school students' perception of the difficulty of Chinese Sign Language (CSL), and analyzes the suggestions proposed by participants. Through a questionnaire survey, data on sign language awareness were collected from 184 secondary school students. The results indicate that although most students had been exposed to sign language through online platforms, public welfare activities, and other channels, their perception of its difficulty shifted noticeably after exposure to basic sign language linguistics. Notably, students commonly reported that their unfamiliarity with sign language stemmed primarily from a lack of formal learning platforms.

Keywords: Sign Language Linguistics、 Perception of Difficulty、 Language Cognition、 Survey Research 、 Sign Formation Mechanisms、 Language Acquisition

CHAPTER 1

Introduction

Chinese Sign Language (CSL) serves as the primary means of communication for the hearing-impaired community and is recognized as a natural language with an independent vocabulary and grammatical system. In September 2025, the Standing Committee of the National People's Congress deliberated the draft revision of the National Common Language and Script Law, during which multiple committee members proposed the inclusion of National Common Sign Language and Braille under legal protection. Concurrently, the National Occupational Standards for Sign Language Interpreters (2025 Edition) were formally promulgated. Despite these policy advancements, the popularization of sign language continues to face considerable challenges, particularly among adolescent populations, whose understanding and awareness of sign language remain limited.

Secondary school students are at a critical stage of cognitive development, during which their language attitudes and learning dispositions exert a significant influence on the broader sociolinguistic landscape. However, most secondary school students' exposure to sign language remains confined to fragmented and often trivialized content—such as a handful of commonly known lexical items (e.g., "thank you") encountered through short videos, films, or television programs—while their understanding of the everyday vocabulary essential to sign language users remains vague and underdeveloped.

Although a substantial body of research has examined language acquisition and social inclusion among the hearing-impaired population, relatively few studies have investigated how knowledge of sign language linguistics shapes the perceptions and attitudes of hearing secondary school students toward sign language. Existing sign language promotion efforts have largely focused on policy advocacy and public

welfare campaigns, with insufficient attention devoted to cognitive-level interventions. In other words, it remains unclear whether—after being introduced to the fundamental principles of sign language word formation—secondary school students can infer the meanings of unfamiliar signed expressions, and thereby offer basic assistance to sign language users in real-world contexts.

To address this research gap, the present study departs from grand narratives and focuses specifically on an empirical analysis of Chinese secondary school students' awareness of Chinese Sign Language, with participants drawn primarily from the middle and high school divisions of Shijiazhuang Foreign Language School.

The study aims to investigate two core questions: (1) Which types of sign language word formation mechanisms are more intuitively accessible to secondary school students? and (2) How does linguistic knowledge influence students' ability to process and interpret signed language? This research aspires to contribute to the broader field of communication between sign language users and non-signers, and its findings may serve as a foundational reference for future investigations. Like other studies in this domain, this work seeks to add valuable empirical insights to the existing literature.

CHAPTER 2

Literature Review

2.1 What is Sign Language Linguistics?

Sign Language Linguistics is an important branch of linguistics that emerged in the 1960s. Drawing on natural sign language as its primary data source, and operating within the theoretical framework of linguistics, it systematically examines the universals and particulars of sign language across multiple levels, including phonology, morphology, syntax, and semantics (Deng, 2003).

The foundational premise of sign language linguistics is that sign language, like spoken language, is a natural language with a complete linguistic system. Sign language is not merely a collection of "gestures" or "motions," but rather a visual language with its own independent vocabulary, grammatical rules, and syntactic structures. As articulated in *The Linguistic Status of Chinese Sign Language (I)* (2009), Chinese Sign Language is an independent visual language that constitutes a linguistic system distinct from Chinese. For deaf individuals, sign language is acquired as their "first language" in the most natural settings, whereas spoken and written language require specialized training. This assertion affirms, from a linguistic perspective, the legitimate status of Chinese Sign Language as a natural language, and provides a theoretical foundation for subsequent sign language instruction, popularization, and linguistic research.

With respect to the internal structure of sign language, Chinese Sign Language adheres to the general principles of linguistics. Research has indicated that the morphological units of Chinese Sign Language are composed of meaningless features, which function analogously to "phonemes" in spoken language—that is, the smallest units capable of distinguishing meaning. Like other human languages, Chinese Sign Language exhibits both arbitrariness and iconicity in its form-meaning relationships

(Phonological Studies of Chinese Sign Language, 2010). This finding suggests that although sign language differs from spoken language in its modality of expression, it remains highly consistent with human language in its organizational principles.

Regarding the specific manifestations of "similarity" (i.e., iconicity) in sign language, A Corpus-Based Study of Iconicity in Chinese Sign Language (2010) further summarizes seven types of iconic devices employed in Chinese Sign Language. These include: direct representation, using the hand to represent the shape of the referent, using the hand to represent the action of the referent, using the body to represent the action of the referent, borrowing from other linguistic systems (such as Chinese characters or Pinyin), and using speech-associated gestures. This classification directly corresponds to the types of sign language word formation examined in the questionnaire of the present study—namely, imitation (iconicity), deixis (pointing), and borrowing—thereby providing theoretical support for the study's questionnaire design.

In recent years, Chinese scholars have begun to investigate the influence of iconicity on sign language learning among hearing populations. The findings suggest that iconicity, as a linguistic feature, plays a facilitative role in the sign language learning process of hearing individuals and provide important theoretical context and empirical grounding for the central question of the present study: whether linguistic knowledge can influence secondary school students' perception and perceived difficulty of sign language.

In summary, research in sign language linguistics has provided substantial evidence for the status of sign language as an independent natural language, and the facilitative role of iconic features and linguistic knowledge in sign language learning has increasingly attracted empirical attention. However, existing studies have predominantly focused on university student populations, leaving the sign language awareness of secondary school students—a critical age group—systematically under-

investigated. The present study is situated precisely within this research gap.

2.2 The role of online platforms in the dissemination of sign language

Since the advent of the internet era, online platforms have gradually emerged as pivotal arenas for the dissemination of sign language. In the realm of online education, Bai and Xu (2026) took sign language courses on the Chinese University MOOC platform as a case study, noting that while online open courses have promoted the sharing of educational resources and enabled broader public access to systematic learning, they also revealed a practical challenge: despite extensive course coverage, learner completion rates remain suboptimal. From the perspective of platform construction, Zhang et al. (2016) investigated how "Internet Plus" sign language teaching platforms can dismantle communication barriers between deaf and hearing individuals. Collectively, these studies indicate that while online platforms have effectively expanded the reach of sign language, there is still significant room for improvement regarding learning depth and systematic instruction.

Concurrently, the lag in the informatization of sign language itself constitutes a more profound constraint. Ren et al. (2023) pointed out that the informatization of Chinese Sign Language significantly trails behind spoken language, manifesting as a lack of theoretical frameworks, practical models, and application software, as well as inadequate sign language datasets and corpora. Although Zhao and Han (2020) proposed new media promotion pathways, such as establishing sign language databases and launching open online courses, a gap remains between these macro-level conceptualizations and the practical dilemma faced by high school students, who often struggle to access reliable learning pathways.

In recent years, the proliferation of short-video and live-streaming platforms has opened new avenues for sign language dissemination. Shen and Guo (2025) found that individuals with disabilities have achieved three levels of digital practice through short-video platforms: "access—connection—output." Other studies have analyzed the media

participation of deaf individuals in Taobao's sign language live-streaming. While these emerging platforms offer new possibilities for dissemination, their content is predominantly characterized by individualized, daily sharing. Lacking pedagogical systematicity and standardization, these platforms create a tension with high school students' need to transition from fragmented exposure to systematic learning.

In conclusion, although online platforms have succeeded in making sign language "visible," the core issue hindering high school students' understanding of sign language is not a lack of visibility, but rather the difficulty in accessing reliable learning pathways that provide substantive support. Further exploration is warranted on how to transition online platforms from merely "expanding reach" to "providing systematic learning pathways."

2.3 Research evidence on the cognitive impacts of linguistic on students

Investigating the cognitive interface between sign and spoken languages, Cao and Li (2021) employed a lexical decision task under priming conditions to examine how iconicity and sign language experience influence cross-modal semantic priming effects between Chinese Sign Language (CSL) and Mandarin Chinese. In their experiment, two groups of participants—fluent signers (CSL-Chinese bilinguals) and hearing non-signers (Chinese monolinguals)—were first presented with video clips of CSL signs, followed immediately by a task to judge whether the subsequent string of Chinese characters constituted a real word.

The study yielded distinct findings based on the nature of the signs. In the iconic condition, both groups demonstrated significant semantic priming effects. However, in the non-iconic condition, significant priming effects were observed exclusively among fluent signers, while hearing non-signers showed no significant effect. These results indicate that while a cross-modal semantic priming effect exists between CSL and Chinese, its emergence is dually modulated by the iconicity of the sign vocabulary and the individual's sign language learning experience (Cao & Li, 2021).

CHAPTER 3

Methodology

3.1 Research Questions

This study investigates secondary school students' perceptions of sign language and their views on learning it. Through four hand sign recognition tasks grounded in basic linguistic knowledge, the study aims to address two core research questions:

1. How does secondary school students' perceived difficulty of sign language change after their exposure to basic linguistic knowledge of sign language?
2. How can the linguistic knowledge that is most intuitively accessible to secondary school students be leveraged to enhance their understanding of sign language?

3.2 Method of Data Collection

This study employed a self-designed online questionnaire as the primary instrument for data collection. The questionnaire was developed and administered through Wenjuanxing (问卷星), a widely used online survey platform in China. It consisted of 10 items and was organized into four sections, each addressing a distinct dimension of the research questions.

Section 1 (Items 1–2) collected demographic information and participants' prior exposure to sign language, including their grade level and the channels through which they had encountered sign language (e.g., online platforms, public welfare activities, school curricula).

Section 2 (Items 3–4) measured participants' perceived difficulty of learning sign language and their attributions of that difficulty. Item 3 asked participants to identify the primary obstacles to learning sign language (e.g., lack of formal channels, limited practical use, perceived complexity), while Item 4 required them to rate the overall difficulty of sign language on a five-point scale (ranging from "very easy" to "very difficult").

Section 3 (Items 5–8) assessed participants' ability to recognize sign language words that exemplify three fundamental types of sign formation: iconicity (mimetic representation), deixis (pointing), and borrowing (from written Chinese or Pinyin). The four items corresponded to the following sign words:

Item	Sign Word	Formation Type	Description
5	Thank you	Iconicity	Mimicking a bowing gesture
6	Train	Iconicity	Simulating a train on the track
7	Ear	Deixis	Pointing directly to the body part
8	Toilet	Borrowing	Borrowed from “W.C.”

(Prior to Item 8, participants were provided with a Chinese Sign Language fingerspelling chart to ensure they possessed the basic information necessary for identifying a borrowing-based sign.)

Section 4 (Items 9–10) repeated the difficulty perception measure from Item 4 to capture any change in perceived difficulty after participants had engaged with the four sign recognition tasks. Item 10 was an open-ended question inviting participants to

share their suggestions, reflections, or additional thoughts regarding sign language learning. This item was optional and served to collect qualitative data to complement the quantitative findings.

The questionnaire was distributed online through multiple channels, including WeChat Moments and the school's confession wall (a student-run social media platform for announcements and peer interaction), between August 15, 2026 and August 18, 2026. All items except Item 10 were mandatory, and answers were automatically recorded by the survey platform.

3.3 Research Participants

The participants in this study were 184 Chinese secondary school students, the majority of whom were recruited from Shijiazhuang Foreign Language School. The sample consisted of 58 middle school students, 112 high school students, and a negligible number of 14 participants from other educational backgrounds. Over 98% of the participants reported having had no prior exposure to formal linguistic knowledge. This background characteristic made it possible to examine the influence of basic linguistic knowledge on secondary school students' understanding of sign language without the confounding effect of prior training.

3.4 Data analysis

Data analysis in this study consisted of the following steps.

First, all closed-ended responses were exported from Wenjuanxing to Microsoft Excel, where the distribution of responses for each item was calculated and tabulated. This step provided an overview of participants' demographic information, prior exposure to sign language, perceived difficulty, and performance on the four sign-recognition tasks.

Second, a pre-post comparison was conducted between Item 4 (pre-test, measuring

participants' perceived difficulty before the sign-recognition tasks) and Item 9 (post-test, measuring the same construct after the tasks). This step was critical for examining whether exposure to basic linguistic knowledge of sign language influenced participants' perception of its difficulty. In addition, cross-tabulations were performed to examine whether participants' prior exposure to sign language or their perception of sign language was associated with their performance on the sign-recognition tasks.

Third, correct rates across the four sign-recognition items (Items 5–8) were compared to identify which of the three sign formation types—iconicity (Items 5 and 6), deixis (Item 7), or borrowing (Item 8)—was most accurately recognized by participants. This step allowed the researcher to determine whether certain types of linguistic knowledge were more intuitively accessible to secondary school students than others.

Finally, responses to the open-ended question (Item 10) were coded thematically. Recurring patterns and representative comments were extracted to complement the quantitative findings.

No advanced statistical software was used; all analyses were conducted using the built-in analytical functions of the survey platform and Microsoft Excel, which were sufficient for the descriptive and comparative purposes of this study.

3.5 Ethical Considerations

All participants took part in this study voluntarily. Prior to participation, they were informed of the purpose of the research, the anonymous nature of the data collection, and their right to withdraw at any time without consequence. Appropriate measures were taken to protect participants' privacy throughout the research process. To ensure anonymity, participants' real names were replaced with numerical codes ranging from 1 to 184 (assigned in the order in which the questionnaires were submitted). No personal identifiable information—such as names, student ID numbers, or class affiliations—was collected or disclosed in any form. All data were

used solely for academic purposes and were accessible only to the researcher.

All data were stored securely on the researcher's password-protected device and will be retained for future academic use, with access restricted to the researcher.

CHAPTER 4

Findings

4.1 The Constraints of Digital Media in Sign Language Dissemination

Table 1 below presents a comparison between participants who had encountered sign language exclusively through digital media (n=85) and those who had encountered it exclusively through public welfare activities (n=6), in terms of their perceived difficulty and performance on the four sign-recognition tasks.

Table 1. Comparison of Sign Recognition Performance by Exposure Channel

Channel	N	"Thank You"	"Train"	"Ear"	"Toilet (W.C.)"	Perceived Difficulty (%Moderate + Difficult)
Digital Media Only	85	82%	49%	100%	49%	83%
Public Welfare Only	6	83%	33%	100%	100%	83%

As shown in the table, the channel of exposure appeared to have little impact on participants' overall perceived difficulty of sign language. In both groups, 83% rated sign language as either "moderate" or "difficult." This suggests that mere exposure—whether through digital media or public welfare activities—does not necessarily reduce the perceived difficulty of learning sign language.

However, notable differences emerged in participants' actual performance on the sign-recognition tasks. The sign "thank you," a basic and highly frequent sign in everyday social interaction, was recognized at similarly high rates across both groups (82% in the digital-media-only group and 83% in the public-welfare-only group). This indicates that the most fundamental and socially visible signs have achieved a relatively high level of popularization among secondary school students, regardless of their primary exposure channel.

The most striking difference lies in the sign "toilet," which is formed through borrowing from the "W.C." While all six participants in the public-welfare-only group correctly identified this sign (100%), fewer than half of the digital-media-only participants (49%) did so. This disparity is particularly significant when considering the practical context: in real-world scenarios where a non-signer might encounter a sign language user in need of assistance, the word "toilet" is far more likely to be relevant than a greeting such as "thank you." The fact that a sign of high practical utility remains unrecognized by over half of digitally-exposed participants points to a critical limitation of current digital media dissemination: while online platforms are effective at popularizing visually intuitive or socially visible signs, they are far less capable of conveying signs that require even a minimal level of linguistic explanation, such as borrowing from "W.C.".

Why not try something more creative—like incorporating current memes into Douyin videos? Lots of people would probably watch them. Or shoot some short skits. If possible, organize a themed class meeting, or use a self-study period to come into class and explain things. (Participant 167)

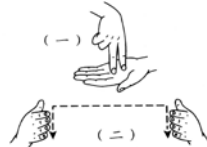
I hope there can be more promotion encouraging everyone to learn more about sign language—for example, by posting promotional videos on online platforms, so that the lives of people with disabilities can be made more convenient. (Participant 156)

4.2 The Logical Aptitude of High School Students for Sign Language Acquisition

Q5 Options	Number of Responses	Percentage
 (Correct Answer: "Thank you")	155	84.24%
	5	2.72%
	4	2.17%
	20	10.87%

In the sign recognition task for "thank you," 84.24% of secondary school students answered correctly. Among the three distractors, Option 4 ("love") received a significantly higher selection rate than Options 2 and 3. This is likely because the gesture visually resembles everyday body language expressing "like" or "love," leading participants to misassociate it with "thank you." This finding reflects that, without formal training, most students can still perform semantic associations based on visual similarity and social context—they are actively linking unfamiliar gestures to possible meanings, a spontaneous display of what can be described as iconicity reasoning in language cognition.

Q6 Options	Number of Responses	Percentage
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	18	9.78%
	50	27.17%
 (Correct Answer: "Train")	96	52.17%
	20	10.87%

In the sign recognition task for "train," 52.17% of secondary school students selected the correct answer. Among the distractors, Option 4 ("platform") received the lowest selection rate, while Option 1 ("tram") was chosen less frequently than Option 2 ("underground"). This pattern suggests that most students understood the hand movement as mimicking the motion of train wheels on tracks. Notably, the stronger interference from semantically related options such as "tram" and "subway"—both within the rail transport category—compared to the visually distinct "platform" indicates that students relied more on semantic relevance than on purely visual similarity. This finding reflects an early manifestation of sign language comprehension from the perspective of embodied cognition.

Q7 Options	Number of Responses	Percentage
 (Correct Answer: "ear")	178	96.74%
	0	0.00%
	5	2.72%
	1	0.54%

In the sign recognition task for "ear," 97% of students answered correctly. This typical deictic sign, formed by pointing to the body part, relies on fundamental spatial cognition and body schema. It is intelligible without prior linguistic knowledge. The near-perfect accuracy is therefore not coincidental. This finding underscores that deixis provides a shared bodily-based cognitive foundation between signers and non-signers, facilitating natural cross-group communication.

Q8 Options	Number of Responses	Percentage
	29	15.76%
 (Correct Answer: "toilet")	98	53.26%

	39	21.20%
	18	9.78%

In the sign recognition task for "toilet," 53% of students selected the correct answer after reviewing the reference information (see Appendix B). Among the distractors, Option 1 ("name/cat") was chosen by 29 participants, likely due to its hand-shape similarity to the correct sign; Option 3 ("small") had a selection rate of 21.20%, possibly because the inward finger movement was associated with the concept of "urgency" and thus contextually linked to "toilet." Overall, with fingerspelling cues provided beforehand, a considerable proportion of students recognized that the sign for "toilet" is formed through a borrowing mechanism from "W.C." This suggests that even for borrowing-based signs lacking direct visual iconicity, secondary school students can grasp their word-formation logic when given appropriate linguistic prompts.

Sign language is a very warm language. I hope schools can offer simple mini-classes on sign language, so that we can learn basic gestures, better understand our hearing-impaired friends, and bridge the gap between us. (Participant 142)

Sign language is very important—it is a means of communication for the hearing-impaired community. I believe we should respect sign language and respect hearing-impaired people, and regard it as a language just like Chinese and English. I think schools or public venues could increase their promotion of sign language, so that everyone can show more respect for sign language and more kindness and understanding toward hearing-impaired people. (Participant 182)

4.3 An Overall Increase in Students' Acceptance of Sign Language Following the Acquisition of Foundational Linguistic Knowledge

Prior to completing the four sign-recognition tasks, participants reported their perceived difficulty of sign language through Item 4 (see Figure 1). After being exposed to basic linguistic knowledge of sign language through the four tasks, participants answered Item 9, which was identical in wording to Item 4 (see Figure 2).

Figure 1

Option	Number of Responses	Percentage
Very difficult — overwhelming at first sight	17	9.24%
Difficult, but manageable for basic greetings	74	40.22%
Moderate — similar to learning a foreign language	71	38.59%
Easy — even interesting	21	11.41%
Very easy — I would like to try incorporating it into daily conversations as a gesture supplement	1	0.54%

Figure2

Option	Number of Responses	Percentage
Very difficult — overwhelming at first sight	20	10.87%

Difficult, but manageable for basic greetings	60	32.61%
Moderate — similar to learning a foreign language	72	39.13%
Easy — even interesting	27	14.67%
Very easy — I would like to try incorporating it into daily conversations as a gesture supplement	5	2.72%

After completing the four sign-recognition tasks, participants' perceived difficulty of sign language was reassessed using the same five-point scale as in Item 4 (see Figures 1 and 2). As shown in Table 4, the proportion of participants who selected "Very difficult — overwhelming at first sight" remained largely unchanged. The "Difficult, but manageable for basic greetings" category decreased by 14 participants. The "Moderate — similar to learning a foreign language" category showed little change. Meanwhile, the "Easy — even interesting" and "Very easy — I would like to try incorporating it into daily conversations as a gesture supplement" categories together increased by 10 participants.

Overall, after engaging with authentic sign language content, participants demonstrated a noticeable reduction in unfamiliarity and perceived difficulty, accompanied by a moderate increase in interest and confidence in learning sign language.

I suggest that primary and secondary schools as well as universities establish sign language interest clubs, and organize activities in the form of games and sign language performances to songs, with a focus on experiencing sign language as a means of

communication. Popularization should not only involve hearing people learning sign language, but also encourage hearing-impaired individuals to learn about mainstream social practices. Offline interactive activities could be created to reduce the gap between the two communities. (Participant 153)

4.4 The Demand for Credible and Formalized Learning Channels Among Non-Signers

Participants' responses indicate that the core challenge in sign language popularization has shifted from raising public awareness of its existence and users to providing non-signers with a publicly accessible, credible, and formalized learning pathway. This recurring demand reflects a broader transition in public expectations—from cognitive awareness to actionable practice. Participants not only wish to be informed about sign language; more fundamentally, they seek systematic learning opportunities that would enable them to acquire basic communicative competence in sign language.

A deaf-mute vendor used to set up a stall near my community. After getting to know him, I learned a few basic signs—thank you, "hello," "goodbye"—and seeing his smile made it all worthwhile. Recently, I have also been learning sign language dance performances and try to make my gestures as accurate as possible before posting them online. It feels very meaningful. (Participant151)

Hello senior. If you are promoting sign language, I have a suggestion: create a set of sign language emojis using frequently used words, with each sign accompanied by corresponding text explanations. Emojis spread easily and could have a strong impact. (Participant 179)

High schools could offer sign language clubs and seek guidance from professionals

while also researching market demand. For middle school social practice programs, consideration could be given to helping students from special education schools integrate into mainstream society. (Participant 171)

CHAPTER 5

Conclusion

This study aims to explore an effective pathway enabling non-signers to comprehend basic information expressed by signers in specific contexts. The findings reveal that after acquiring foundational word-formation knowledge in sign language linguistics, high school students exhibited a significant reduction in apprehension toward sign language, accompanied by an overall enhancement in their learning confidence. Furthermore, the results indicate that among the three word-formation mechanisms—deixis, imitation, and borrowing—deictic signs are the most intuitive and comprehensible for non-signers.

The conclusions of this study provide a valuable supplement to the existing literature. First, it reveals that non-signers possess an intuitive capacity for sign language comprehension. This finding offers practical implications for daily communication between signers and non-signers, underscoring the critical role of deictic vocabulary in cross-group interactions. Second, the study gathered innovative suggestions from high school students regarding sign language dissemination, providing a youth-oriented perspective for optimizing the promotion pathways of sign language.

Despite these contributions, several limitations must be acknowledged. First, the majority of participants were recruited from a single school and generally demonstrated strong English proficiency. This may introduce a bias toward higher language cognitive abilities within the sample, potentially affecting the generalizability of the results. Second, the study included only four sign language recognition items. The limited number of test items may be insufficient to conclusively guarantee that the accuracy rates were entirely attributable to participants' understanding of the three word-formation mechanisms (deixis, imitation, and borrowing).

Future research could further explore innovative formats for promoting sign language among younger demographics, such as sign language emojis and sign language gesture dances. Integrating systematic sign language linguistics into fragmented, bite-sized science communication content holds the potential to stimulate broader interest and cognitive awareness of sign language across wider populations.

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Appendix

			
A	B	C	D
			
E	F	G	H
			
I	J	K	L
			
M	N	O	P
			
Q	R	S	T
			
U	V	W	X
			
Y	Z	ZH	CH
			
SH	NG	Ê	Ü